**COURSE TITLE : POLYMER SCIENCE LAB**
**COURSE CODE : 2099**
**COURSE CATEGORY : B**
**PERIODS/WEEK : 3**
**PERIODS/SEMESTER: 45/2**
**CREDITS : 2**

## TIME SCHEDULE

| Module | Topics | Period |
|---|---|---|
| 1 | Determination normality and estimation | 15 |
| 2 | Preparation of polymers by different polymerization techniques. | 15 |
| 3 | Determination of the melting point and specific gravity of polymers and rubber chemicals | 15 |
| TOTAL | | 45 |

## COURSE OUTCOME

| Sl. | GO | Student will be able to |
|---|---|---|
| 1 | 1 | To find out the normality and weight per litre of solute of solutions by titration methods |
| 2 | 2 | To prepare polymers by different polymerization techniques |
| 3 | 3 | To determine the melting point of polymers and rubber chemicals by thiels tube method |
| 4 | 4 | To determine the specific gravity of plastics and elastomers |

## DETAILS OF LABORATORY WORK IN POLYMER SCIENCE LAB

### MODULE I DETERMININATION NORMALITY AND ESTIMATION

1.1.0 To find out the normality and weight per litre of solute of solutions by titration methods

1.1.1.Determine the normality of given hydrochloric acid using a standard solution of sodium carbonate solution.

1.1.2. Determine the normality of given NaOH using a standard solution of $H_2C_2O_4$

1.1.3. Determine the strength of given sodium hydroxide solution,provided with a sodium carbonate using a link solution of hydrochloric acid and estimate the amount of NaOH in a given volume

1.1.4. Determine the strength of given oxalic acid solution provided with a solution of HCl and NaOH.

### MODULE II PREPARATION OF POLYMERS BY DIFFERENT POLYMERIZATION TECHNIQUES

2.1.0 To prepare polymers by different polymerization techniques

2.1.1.Carry out the preparation of UF resin

2.1.2.Carry out the preparation of PF resin

2.1.3.Carry out the preparation of PMMA

2.1.4.Carry out the preparation of Regenerated cellulose

## MODULE III DETERMINATION OF THE MELTING POINT AND SPECIFIC GRAVITY OF POLYMERS AND RUBBER CHEMICALS

3.1.0 To determine the specific gravity of plastics and elastomers

- 3.1.1. Determine the melting point of wax.
- 3.1.2. Determine the melting point of Stearic acid.
- 3.1.3. Determine the melting point of polypropylene.
- 3.1.4. Determine the melting point of polyethylene
- 3.1.5. Find out the specific gravity of polymer sample by Hydrostatic bench method
- 3.1.6. Find out the specific gravity of polymer sample by bottle method